IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday, 1 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. A rigid body is fixed on Earth at its centre of mass and rotates about the fixed point. The origin of the body-fixed axes is at the fixed point and the body-fixed axes are principal axes. I_{xx}, I_{yy}, and I_{zz} are the three principal moments of inertia of the rigid body with respect to the body-fixed axes, and Nx, N_y, and N_z are the three components of the moment acting on the rigid body along the body-fixed axes, respectively. The angular velocity vector of the rigid body in the body-fixed coordinate system is

$$\boldsymbol{\Omega} = \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix}$$

   a) Write the Euler equations of motion for the rigid body with one point fixed. [3]

   b) The rotation of the rigid body with respect to Earth-fixed axes is represented by three Euler angles ψ, θ and ϕ in the yaw-pitch-roll convention. In this convention the body is first rotated from its nominal configuration by an angle ψ about the z-axis, then by an angle θ about the intermediate y-axis of the body and finally by an angle ϕ about the new x-axis of the body. By making use of the standard single-axis-rotation transformation matrices, calculate Ω_x, Ω_y, and Ω_z in terms of the generalised coordinates, ψ, θ and ϕ, and their derivatives. [4]

   c) Compute the Lagrangian function of the rigid body. [5]

   d) Derive one of the Euler equations of motion from the Lagrangian equation of motion for the ϕ generalised coordinate. [8]

2. A gantry robotic arm consists of a cart of mass M and a uniform arm of length l, attached on the cart at point O, as shown in Figure 2.1, with any interaction between the cart and the arm occurring only at the joint O. The cart translates on a smooth (frictionless) horizontal surface by a distance x, under the influence of a cart-generated horizontal control force F_c (acting from Earth on the cart) and the interaction with the arm. The arm is free to rotate in a vertical plane about a normal-to-the-diagram axis through O by an angle θ, under the influence of gravity, a control torque T_a acting from the cart on the arm, and the interaction with the cart. The arm mass and moment of inertia about a normal-to-the-diagram axis through its centre of mass is m and I, respectively.

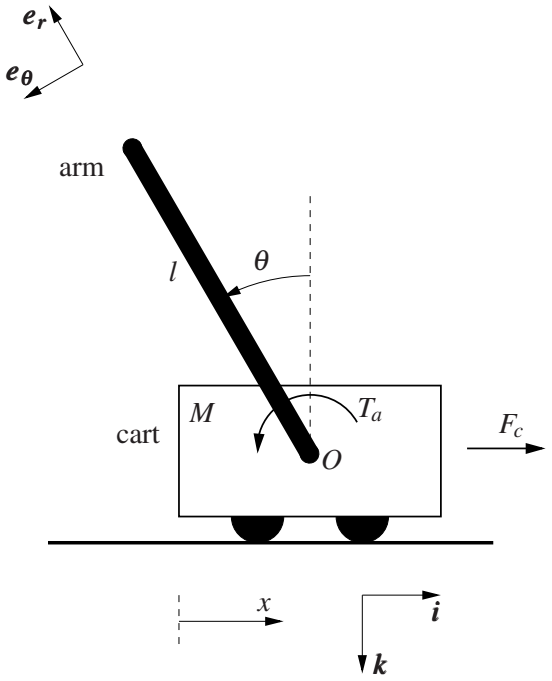


Figure 2.1 Gantry robotic arm with one link.

An Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$, and a moving Cartesian coordinate system with unit vectors $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ are used to analyse the motion of the gantry robotic arm. The moving coordinate system is attached to the arm, it has its origin at O and it rotates by an angle θ.

a) Write the position vector of the cart and of the arm centre of mass. [3]

b) Calculate the velocity vector of the cart and of the arm centre of mass. [3]

c) Compute the kinetic energy of the system. [3]

d) Compute the potential energy of the system, and hence determine the Lagrangian function. [3]

e) Use the Lagrangian approach to derive the equations of motion of the system in terms of the generalised coordinates. [8]

3. As in Q2, a gantry robotic arm consists of a cart of mass M and a uniform arm of length l, attached on the cart at point O, as shown in Figure 3.1, with any interaction between the cart and the arm occurring only at the joint O. The cart translates on a smooth (frictionless) horizontal surface by a distance x, under the influence of a cart-generated horizontal control force F_c (acting from Earth on the cart) and the interaction with the arm. The arm is free to rotate in a vertical plane about a normal-to-the-diagram axis through O by an angle θ, under the influence of gravity, a control torque T_a acting from the cart on the arm, and the interaction with the cart. The arm mass and moment of inertia about a normal-to-the-diagram axis through its centre of mass is m and I, respectively.

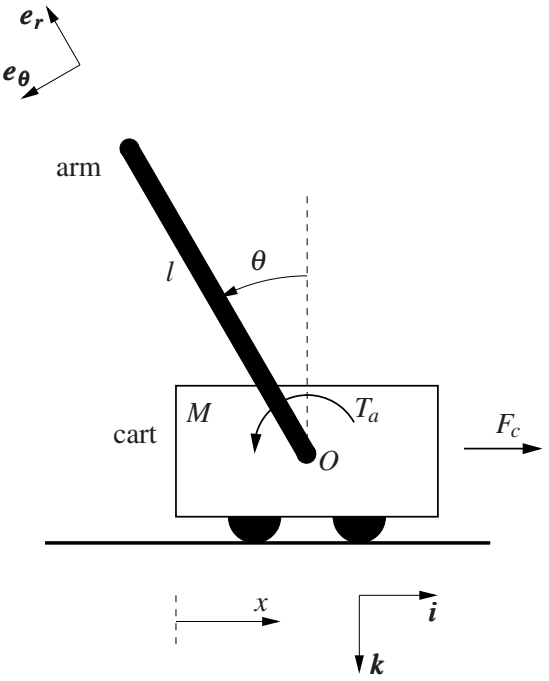


Figure 3.1 Gantry robotic arm with one link.

An Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$, and a moving Cartesian coordinate system with unit vectors $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ are used to analyse the motion of the gantry robotic arm. The moving coordinate system is attached to the arm, it has its origin at O and it rotates by an angle θ.

a) Determine the generalised coordinates of the system. [2]

b) Write the acceleration vector of the cart and the arm centre of mass. [3]

c) F_x and F_z are the unknown constraint forces on the arm from the cart in the $\boldsymbol{i}$ and $\boldsymbol{k}$ directions, respectively. Write the total force vector that acts, respectively, on the cart and the arm. [3]

d) Use the vectorial approach to derive F_x and F_z in terms of the generalised coordinates. [5]

e) Use the vectorial approach to further derive the equations of motion of the system with respect to the generalised coordinates, and the control force and torque. [7]

4. Consider the position vector, $\boldsymbol{r_i}$, with respect to a fixed origin O, of the ith mass particle of mass m_i of a rigid body moving in three-dimensional space, in relation to the rigid body centre of mass location, C, or the location of another point, A, attached to the rigid body, as shown in Figure 4.1. The total mass of the rigid body is $M = \sum_i m_i$ and its angular velocity vector is $\boldsymbol{\Omega}$.

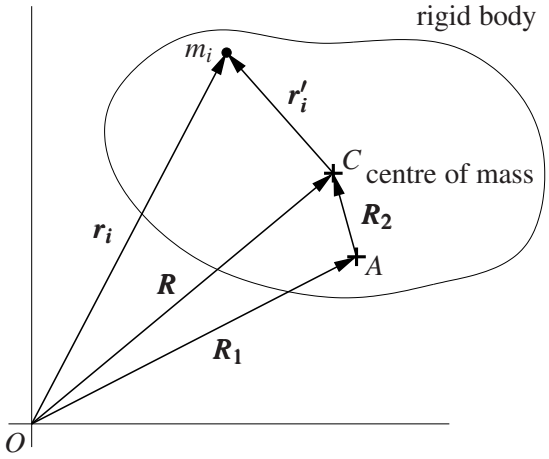


Figure 4.1 Vectors involved in the calculation of various quantities related to the motion of a rigid body.

a) Write the relationship between the position vectors

i) $\boldsymbol{r_i}$, $\boldsymbol{R}$, and $\boldsymbol{r'_i}$,

ii) $\boldsymbol{R}$, $\boldsymbol{R_1}$, and $\boldsymbol{R_2}$, and

iii) $\boldsymbol{r_i}$, $\boldsymbol{R_1}$, $\boldsymbol{R_2}$, and $\boldsymbol{r'_i}$.

[3]

b) Calculate the velocity vector $\frac{d\boldsymbol{r_i}}{dt} = \boldsymbol{v_i}$ of the ith mass particle of the rigid body by involving

i) the velocity vector of the centre of mass, C, and

ii) the velocity vector of point A and of the centre of mass, C.

[4]

c) The total kinetic energy of the rigid body motion about point O is from first principles the sum of the kinetic energy of all mass particles, $T = \sum_i \frac{1}{2} m_i v_i^2$, and it can be shown, as in the notes, to split neatly into one term relating to the translational motion of the centre of mass and another involving rotation about the centre of mass. Instead of such a split, compute and simplify the kinetic energy of the rigid body using point A (make use of the result from part b) ii)).

[5]

d) As in the notes, it can be shown from first principles that:

- The translational motion of the rigid body is defined by the motion of the centre of mass as if all the mass was concentrated at that point, as follows

$$M\frac{d^2\boldsymbol{R}}{dt^2} = \sum_i \boldsymbol{F_i^{(e)}} \equiv \boldsymbol{F}^{(e)},$$

where $\boldsymbol{F}_i^{(e)}$ is the external force acting on the ith mass particle and $\boldsymbol{F}^{(e)}$ is the total external force acting on the rigid body.

- The total angular momentum vector of the rigid body about the fixed point O, $\boldsymbol{H} = \sum_i \boldsymbol{r_i} \times \boldsymbol{p_i}$, where $\boldsymbol{p_i} = m_i \frac{d\boldsymbol{r_i}}{dt}$ is the linear momentum of the ith mass particle, divides naturally into two contributions, one from the translational motion of the centre of mass (as if all the mass of the rigid body is concentrated at the centre of mass) and one from the rotational motion about the centre of mass, as follows

$$\boldsymbol{H} = \boldsymbol{R} \times M\frac{d\boldsymbol{R}}{dt} + \boldsymbol{H_0}, \tag{4.1}$$

  where $\boldsymbol{H_0} = \sum_i \boldsymbol{r_i'} \times m_i \frac{d\boldsymbol{r_i'}}{dt}$ is the angular momentum of motion about the centre of mass, with $\frac{d\boldsymbol{r_i'}}{dt}$ the velocity of the ith particle relative to the centre of mass.
- The rotational motion of the rigid body about fixed point O is given by

$$\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}^{(e)}, \tag{4.2}$$

  with the total moment about O of external forces acting on the rigid body

$$\boldsymbol{N}^{(e)} = \sum_i \boldsymbol{r_i} \times \boldsymbol{F_i}^{(e)},$$

  and from (4.1)

$$\frac{d\boldsymbol{H}}{dt} = \boldsymbol{R} \times M\frac{d^2\boldsymbol{R}}{dt^2} + \frac{d\boldsymbol{H_0}}{dt},$$

  which also splits neatly into one term relating to the translational motion of the centre of mass and another involving rotation about the centre of mass.

Derive an equation, analogous to (4.2), for the rotational motion of the rigid body about point A, which still involves $\boldsymbol{H_0}$ and where the right hand side of the equation is $\boldsymbol{N}_A^{(e)}$, the total moment about point A of external forces acting on the rigid body. [8]